

MATH-2111

Tutorial 2: Linear Independence, Linear Transformations & Standard Matrices

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Conceptual Review I: Existence vs. Uniqueness

Let $A = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_p]$. For a coefficient vector $\mathbf{x} = (x_1, \dots, x_p)^T$,

$$A\mathbf{x} = x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p.$$

Existence: a span question

For a target vector $\mathbf{b}$,

$$\mathbf{b} \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$$

exactly when $A\mathbf{x} = \mathbf{b}$ is consistent. The coefficients in a solution tell us how to build $\mathbf{b}$.

Uniqueness: an independence question

If $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, then $\Rightarrow$ Homogeneous system $A\mathbf{z} = \mathbf{0}$
 $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$. only has trivial solution

Thus two coefficient lists differ by a solution of the homogeneous equation.

The central idea

Nonzero solutions of $A\mathbf{z} = \mathbf{0}$ are exactly the **directions of non-uniqueness**. If $A\mathbf{x}_p = \mathbf{b}$, then

$$A(\mathbf{x}_p + t\mathbf{z}) = \mathbf{b} \quad (t \in \mathbb{R}).$$

Conceptual Review II: Four Languages for Independence

For $A = [\mathbf{v}_1 \cdots \mathbf{v}_p]$, the following statements express the same fact.

Language	Meaning of linear independence
Vector equation	$x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$ forces $x_1 = \cdots = x_p = 0$.
Linear system	$A\mathbf{x} = \mathbf{0}$ has no free variables.
Pivot structure	A has a pivot position in every column.
Information / span	Every vector contributes a genuinely new direction; removing any one vector changes the span.

How to choose a viewpoint

Use the **equation** to prove a theorem, **row reduction** to calculate, **pivots** to classify quickly, and **redundancy** to interpret the result.

Conceptual Review III: Why Dependence Means Redundancy

Suppose

$$c_1 \mathbf{v}_1 + \cdots + c_p \mathbf{v}_p = \mathbf{0}$$

for coefficients that are not all zero. Choose an index j with $c_j \neq 0$. Then

$$\mathbf{v}_j = - \sum_{i \neq j} \frac{c_i}{c_j} \mathbf{v}_i$$

so $\mathbf{v}_j$ is built from the other vectors and adds no new direction.

The converse uses the same algebra backwards

If one vector is a linear combination of the others, move all terms to one side. This produces a nontrivial relation equal to $\mathbf{0}$, so the set is linearly dependent.

Quantifier that matters

Dependence guarantees that **at least one** vector is redundant, not that every vector is. For example, in $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{0}\}$ only the zero vector is forced to be redundant.

Conceptual Review IV: The Pivot Budget

Let A be an $m \times p$ matrix whose columns are p vectors in $\mathbb{R}^m$.

Why $p > m$ forces dependence

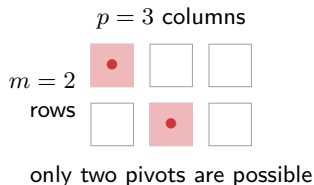
An $m \times p$ matrix has at most m pivots. If $p > m$, at least one column cannot contain a pivot:

$p > m \implies$ at least one nonpivot column

$\implies$ a free variable in $A\mathbf{x} = \mathbf{0}$

$\implies$ a nontrivial solution (only trivial solution exists means linearly independent)

Hence the columns are linearly dependent.



Three familiar rules are all pivot failures

A set containing $\mathbf{0}$ has a zero column; two scalar multiples produce the same direction; more than m vectors in $\mathbb{R}^m$ exceed the pivot budget.

Conceptual Review V: What Linearity Really Preserves

The two axioms of linearity combine into one master identity:

Definition

$$T(c_1 \mathbf{u}_1 + \cdots + c_p \mathbf{u}_p) = c_1 T(\mathbf{u}_1) + \cdots + c_p T(\mathbf{u}_p)$$

Relations

If $\sum c_i \mathbf{u}_i = \mathbf{0}$, then

$$\sum c_i T(\mathbf{u}_i) = \mathbf{0}.$$

Every linear relation among inputs survives in the outputs.

Spans

$$T(\text{Span}\{\mathbf{u}_i\}) = \text{Span}\{T(\mathbf{u}_i)\}.$$

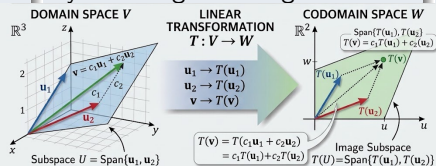
A whole subspace is determined by the images of its generators.

The origin

Since $\mathbf{0} = 0\mathbf{u}$,

$$T(\mathbf{0}) = \mathbf{0}.$$

Lines and planes through the origin remain linear objects, although they may collapse.



Conceptual Review VI: Why Basis Images Determine the Map

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of $\mathbb{R}^n$. Every vector has a unique expansion

$$\mathbf{x} = \alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n.$$

Therefore linearity forces

$$T(\mathbf{x}) = \alpha_1 T(\mathbf{b}_1) + \dots + \alpha_n T(\mathbf{b}_n).$$

Standard basis

Taking $\mathbf{b}_j = \mathbf{e}_j$ gives

$$A = [T(\mathbf{e}_1) \quad \dots \quad T(\mathbf{e}_n)], \quad T(\mathbf{x}) = A\mathbf{x}.$$

The columns of A are not arbitrary: they are the images of the coordinate directions.

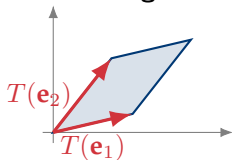
Nonstandard data

Images of any basis also determine T . But if the supplied input vectors are dependent, their images must obey the same relations. For example,

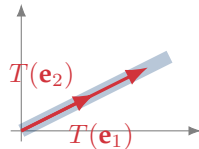
$$\mathbf{b}_3 = \mathbf{b}_1 + \mathbf{b}_2 \implies T(\mathbf{b}_3) = T(\mathbf{b}_1) + T(\mathbf{b}_2).$$

Conceptual Review VII: Geometry Hidden in the Columns

Independent image directions



The unit square becomes a genuine parallelogram.
No coordinate direction is redundant.



Dependent image directions

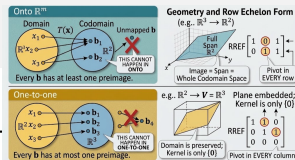
The square collapses to a line. A nonzero input direction is lost, so the map is not one-to-one and its range is not all of $\mathbb{R}^2$.

Geometric reading of the standard matrix

The columns tell us where the coordinate axes go; their span is the range, and their dependence records whether any input direction is collapsed.

Conceptual Review VIII: Onto and One-to-One

For $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $T(\mathbf{x}) = A\mathbf{x}$:



Meaning for $A\mathbf{x} = \mathbf{b}$

Pivot test

Onto $\mathbb{R}^m$

Every $\mathbf{b}$ has at least one preimage.

Pivot in every row.

One-to-one

Every $\mathbf{b}$ has at most one preimage.

Pivot in every column; equivalently $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{0}$.

Existence and uniqueness are separate questions

Onto asks whether a solution exists for *every* target. One-to-one asks whether two different inputs can ever have the same image.

Dimension screen

$n > m$ makes one-to-one impossible; $n < m$ makes onto impossible. When $n = m$, the two properties are equivalent because “pivot in every column” and “pivot in every row” are the same condition.

Conceptual Review IX: The Structure of Every Preimage

Assume $A\mathbf{x}_p = \mathbf{b}$. If $A\mathbf{x} = \mathbf{b}$ as well, then

$$A(\mathbf{x} - \mathbf{x}_p) = \mathbf{0}.$$

Thus $\mathbf{x} - \mathbf{x}_p$ is a homogeneous solution. Conversely, adding any homogeneous solution to $\mathbf{x}_p$ does not change the image.

Universal formula

$$\{\mathbf{x} : A\mathbf{x} = \mathbf{b}\} = \mathbf{x}_p + \{\mathbf{z} : A\mathbf{z} = \mathbf{0}\}.$$

Onto / range

A particular solution $\mathbf{x}_p$ establishes that the target $\mathbf{b}$ can be reached.

One-to-one

The preimage is unique exactly when the only homogeneous direction is $\mathbf{z} = \mathbf{0}$.

Problem 1: A Parameter and Linear Independence

Problem 1

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ h \\ 2 \end{bmatrix}.$$

- (a) For which values of h is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent?
- (b) At the exceptional value of h , find a nontrivial linear relation and identify the redundant vector.
- (c) Explain what changes geometrically at the exceptional value.

Solution to Problem 1

Place the vectors as columns and row reduce:

$$A_h = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & h \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & h - 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & h - 2 \\ 0 & 0 & 4 - h \end{bmatrix}.$$

- If $h \neq 4$, there is a pivot in every column, so the set is **linearly independent**.
- If $h = 4$, take $x_3 = 1$ in $A_4\mathbf{x} = \mathbf{0}$:

$$-2\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}, \quad \boxed{\mathbf{v}_3 = 2\mathbf{v}_1 + 2\mathbf{v}_2}.$$

Geometric interpretation

For $h \neq 4$, the three vectors provide three independent directions and span $\mathbb{R}^3$. At $h = 4$, the third vector falls into the plane spanned by the first two, so the span drops to a plane.

Problem 3: Is the Map Linear?

Problem 3

Determine whether each map is linear. Give a decisive justification.

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2,$

$$T(x_1, x_2) = (2x_1 - x_2, x_1 + 3x_2).$$

If it is linear, find its standard matrix.

(b) $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2,$

$$S(x_1, x_2) = (2x_1 - x_2 + 1, x_1 + 3x_2).$$

(c) $U : \mathbb{R}^2 \rightarrow \mathbb{R},$

$$U(x_1, x_2) = x_1x_2.$$

Solution to Problem 3

- (a) T is linear, because

$$T(\mathbf{x}) = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \mathbf{x}.$$

Hence $A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$.

- (b) S is not linear, since

$$S(\mathbf{0}) = S(0, 0) = (1, 0) \neq (0, 0).$$

- (c) U is not linear. Although $U(\mathbf{0}) = 0$, additivity fails:

$$U(\mathbf{e}_1) + U(\mathbf{e}_2) = 0, \quad U(\mathbf{e}_1 + \mathbf{e}_2) = U(1, 1) = 1.$$

Diagnostic principle

The zero test is an efficient way to disprove linearity, but passing it does not prove linearity. A valid proof must establish preservation of addition and scalar multiplication, or exhibit a matrix formula $T(\mathbf{x}) = A\mathbf{x}$.

Problem 4: Recovering the Standard Matrix from a Different Basis

Problem 4

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be linear and suppose

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = [\vec{a}_1 \ \vec{a}_2]^T \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \quad T \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}.$$

- (a) Explain why these two pieces of information determine T uniquely.
- (b) Find $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$.
- (c) Find the standard matrix and a formula for $T(x_1, x_2)$.
- (d) Compute $T(3, -2)$.

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \therefore \begin{cases} \vec{a}_1 + \vec{a}_2 = \vec{b}_1 \\ \vec{a}_1 - \vec{a}_2 = \vec{b}_2 \end{cases} \Rightarrow \begin{cases} \vec{a}_1 = (\vec{b}_1 + \vec{b}_2)/2 = (1, 2, 0)^T \\ \vec{a}_2 = (\vec{b}_1 - \vec{b}_2)/2 = (1, -1, 2)^T \end{cases}$$

Solution to Problem 4: From the Given Basis to the Standard Basis

Set

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are not scalar multiples, so they are linearly independent and hence form a basis of $\mathbb{R}^2$. Their images therefore determine T uniquely.

Since

$$\mathbf{e}_1 = \frac{1}{2}(\mathbf{u} + \mathbf{v}), \quad \mathbf{e}_2 = \frac{1}{2}(\mathbf{u} - \mathbf{v}),$$

linearity gives

$$T(\mathbf{e}_1) = \frac{1}{2}(T(\mathbf{u}) + T(\mathbf{v})) = \boxed{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}},$$

$$T(\mathbf{e}_2) = \frac{1}{2}(T(\mathbf{u}) - T(\mathbf{v})) = \boxed{\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}}.$$

Solution to Problem 4: Standard Matrix and Evaluation

The standard matrix has columns $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Hence

$$T(x_1, x_2) = \begin{bmatrix} x_1 + x_2 \\ 2x_1 - x_2 \\ x_1 + 2x_2 \end{bmatrix}.$$

For the requested vector,

$$T(3, -2) = \begin{bmatrix} 3 - 2 \\ 6 + 2 \\ 3 - 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -1 \end{bmatrix}.$$

Conceptual point

Information may be given on any basis; the standard matrix is obtained by converting those basis vectors to $\mathbf{e}_1, \mathbf{e}_2$.

Problem 5: Onto, One-to-One, and All Preimages

Problem 5

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation with standard matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{bmatrix}.$$

- (a) Is T onto $\mathbb{R}^2$?
- (b) Is T one-to-one?
- (c) Find every $\mathbf{x} \in \mathbb{R}^3$ such that

$$T(\mathbf{x}) = \begin{bmatrix} 2 \\ 7 \end{bmatrix}.$$

- (d) Interpret the answer using “particular solution + homogeneous solution.”

Solution to Problem 5: Classification

Row reduce the coefficient matrix:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- There is a pivot in every row, so T is onto $\mathbb{R}^2$.
- There is no pivot in column 2, so $A\mathbf{x} = \mathbf{0}$ has a free variable. Thus T is not one-to-one.

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- There is no pivot in column 2, so $A\mathbf{x} = \mathbf{0}$ has a free variable. Thus T is not one-to-one.

For the prescribed image,

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 4 & 1 & 7 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right].$$

Solution to Problem 5: The Entire Preimage

The row-reduced system is

$$x_1 + 2x_2 = 3, \quad x_3 = 1.$$

Let $x_2 = t$. Then

$$\mathbf{x} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad t \in \mathbb{R}.$$

Particular solution

$$\mathbf{x}_p = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \quad A\mathbf{x}_p = \begin{bmatrix} 2 \\ 7 \end{bmatrix}.$$

This establishes existence.

Homogeneous direction

$$\mathbf{z} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad A\mathbf{z} = \mathbf{0}.$$

Adding multiples of $\mathbf{z}$ leaves the image unchanged.

Problem 6: How Linear Relations Move Through a Map

Problem 6

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ have standard matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix},$$

and let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

- (a) Find a relation among $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ and use linearity to predict a relation among their images before multiplying by A .
- (b) Compute the three images and verify the prediction.
- (c) Are $T(\mathbf{u}_1)$ and $T(\mathbf{u}_2)$ linearly independent? Determine $T(\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\})$.
- (d) Which direction of implication between independence of inputs and outputs is always valid? When does the reverse direction hold?

Solution to Problem 6: Preserving Relations and Spans

Since

$$\mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2,$$

linearity predicts

$$\boxed{T(\mathbf{u}_3) = T(\mathbf{u}_1) + T(\mathbf{u}_2)}.$$

Direct calculation gives

$$T(\mathbf{u}_1) = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad T(\mathbf{u}_2) = \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \quad T(\mathbf{u}_3) = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

and indeed $(0, 3)^T = (3, 2)^T + (-3, 1)^T$.

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and indeed $(0, 3)^T = (3, 2)^T + (-3, 1)^T$.

To test the first two images, suppose

$$a \begin{bmatrix} 3 \\ 2 \end{bmatrix} + b \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \mathbf{0}.$$

Then $3a - 3b = 0$ gives $a = b$, while $2a + b = 0$ gives $3a = 0$. Hence $a = b = 0$. Therefore the two images are independent and span $\mathbb{R}^2$:

$$T(\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}) = \mathbb{R}^2.$$

Solution to Problem 6: The Implication That Always Holds

Dependence is preserved by every linear map

If the input vectors are dependent, then for some coefficients not all zero,

$$\sum c_i \mathbf{u}_i = \mathbf{0}.$$

Applying T gives

$$\sum c_i T(\mathbf{u}_i) = T(\mathbf{0}) = \mathbf{0},$$

so the images are dependent with the same coefficients.

Contrapositive

If the images $\{T(\mathbf{u}_i)\}$ are linearly independent, then the original inputs $\{\mathbf{u}_i\}$ must also be linearly independent.

Solution to Problem 6: When Independence Is Preserved

The reverse implication needs injectivity

Independent inputs can become dependent if a linear map collapses a direction. The implication

$$\{\mathbf{u}_i\} \text{ independent} \implies \{T(\mathbf{u}_i)\} \text{ independent}$$

holds when T is one-to-one on $\text{Span}\{\mathbf{u}_i\}$; in particular, it holds when T is one-to-one on its whole domain.

Subtle point in this problem

The map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ cannot be one-to-one on all of $\mathbb{R}^3$. Nevertheless, its restriction to the plane $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is one-to-one because the two images are independent.

Equivalent local test

On a chosen subspace, T is one-to-one exactly when no nonzero vector in that subspace is sent to $\mathbf{0}$.

Problem 7: A Parameterized Linear Transformation

Problem 7

For $h \in \mathbb{R}$, let $T_h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ have standard matrix

$$A_h = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & h \end{bmatrix}.$$

- (a) Determine all h for which T_h is one-to-one and all h for which it is onto.
- (b) When $h = 2$, find an algebraic condition on b_1, b_2, b_3 for $\mathbf{b} = (b_1, b_2, b_3)^T$ to lie in the range of T_2 .
- (c) When $h = 2$, find all preimages of $\mathbf{b} = (2, -1, 1)^T$.
- (d) Connect the homogeneous direction, the column relation, and the geometry of the range.

Solution to Problem 7: One-to-One and Onto

Row reduce A_h :

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & h \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & h - 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & h - 2 \end{bmatrix}.$$

Solution to Problem 7: One-to-One and Onto

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- If $h \neq 2$, there is a pivot in every column and every row. Hence T_h is both one-to-one and onto.
- If $h = 2$, there are only two pivots. Hence T_2 is neither one-to-one nor onto $\mathbb{R}^3$.

Thus

$$\boxed{T_h \text{ is one-to-one} \iff h \neq 2} \quad \text{and} \quad \boxed{T_h \text{ is onto} \iff h \neq 2}.$$

Solution to Problem 7: The Range at $h = 2$

For $h = 2$, reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & b_1 \\ 0 & 1 & 1 & b_2 \\ 1 & 1 & 2 & b_3 \end{array} \right] \xrightarrow{R_3 - R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & b_1 \\ 0 & 1 & 1 & b_2 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array} \right].$$

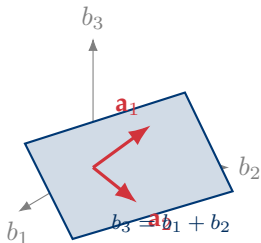
The system is consistent exactly when

$$b_3 = b_1 + b_2.$$

Hence

$$\text{Range}(T_2) = \left\{ \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3 : b_3 = b_1 + b_2 \right\},$$

a plane through the origin.



Solution to Problem 7: All Preimages and the Common Cause

For $\mathbf{b} = (2, -1, 1)^T$, the range condition holds because $1 = 2 + (-1)$. The independent equations are

$$x_1 + x_3 = 2, \quad x_2 + x_3 = -1.$$

Let $x_3 = t$. Then

$$\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

One vector explains three phenomena

The vector $\mathbf{z} = (-1, -1, 1)^T$ satisfies $A_2\mathbf{z} = \mathbf{0}$. Therefore:

- distinct inputs separated by multiples of $\mathbf{z}$ have the same image;
- T_2 is not one-to-one;
- the column relation $-\mathbf{a}_1 - \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$ shows that the range has only two independent directions, namely a plane.

Problem 8: Capstone—Consistent Data, Matrix, and Preimages

Problem 8

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

A linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is claimed to satisfy

$$T(\mathbf{u}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T(\mathbf{u}_2) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(\mathbf{u}_4) = \begin{bmatrix} k \\ 4 \end{bmatrix}.$$

- (a) Determine the value of k that is necessary and sufficient for all four claims to be consistent with linearity.
- (b) For that value of k , find the standard matrix of T .
- (c) Decide whether T is onto and whether it is one-to-one.
- (d) Find all $\mathbf{x}$ satisfying $T(\mathbf{x}) = T(\mathbf{u}_1)$.

Solution to Problem 8: Consistency of the Prescribed Images

The input vectors satisfy

$$\mathbf{u}_4 = \mathbf{u}_1 + \mathbf{u}_2 - 2\mathbf{u}_3.$$

Therefore linearity requires

$$T(\mathbf{u}_4) = T(\mathbf{u}_1) + T(\mathbf{u}_2) - 2T(\mathbf{u}_3) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

Hence

$$\boxed{k = -1}.$$

Solution to Problem 8: Consistency of the Prescribed Images

The input vectors satisfy

$$\mathbf{u}_4 = \mathbf{u}_1 + \mathbf{u}_2 - 2\mathbf{u}_3.$$

Therefore linearity requires

$$T(\mathbf{u}_4) = T(\mathbf{u}_1) + T(\mathbf{u}_2) - 2T(\mathbf{u}_3) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

Hence

$$\boxed{k = -1}.$$

This condition is also sufficient: $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ form a basis because

$$\mathbf{e}_3 = \mathbf{u}_3, \quad \mathbf{e}_1 = \mathbf{u}_1 - \mathbf{u}_3, \quad \mathbf{e}_2 = \mathbf{u}_2 - \mathbf{u}_3.$$

Thus their prescribed images define a unique linear map, and $k = -1$ makes the fourth datum agree with that map.

Solution to Problem 8: Standard Matrix

Using the standard basis expressions,

$$T(\mathbf{e}_1) = T(\mathbf{u}_1) - T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$T(\mathbf{e}_2) = T(\mathbf{u}_2) - T(\mathbf{u}_3) = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad T(\mathbf{e}_3) = T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Therefore

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 0 \end{bmatrix}.$$

Internal check

Multiplying A by each $\mathbf{u}_i$ reproduces all four prescribed images when $k = -1$. This verifies both the basis conversion and the consistency calculation.

Solution to Problem 8: Mapping Properties and All Preimages

Row reduction gives

$$\begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 0 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 5 & -1 \end{bmatrix}.$$

There is a pivot in every row but not every column, so

$$T \text{ is onto } \mathbb{R}^2 \text{ but not one-to-one.}$$

For $T(\mathbf{x}) = T(\mathbf{u}_1) = (2, 1)^T$,

$$x_1 - 2x_2 + x_3 = 2, \quad x_1 + 3x_2 = 1.$$

Let $x_2 = t$. Then $x_1 = 1 - 3t$ and $x_3 = 1 + 5t$, so

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -3 \\ 1 \\ 5 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The particular solution is $\mathbf{u}_1$; the direction $(-3, 1, 5)^T$ solves $A\mathbf{z} = \mathbf{0}$.

Final Concept Map

One matrix organizes all the questions

For $T(\mathbf{x}) = A\mathbf{x}$ with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$:

Question	Decisive test / interpretation
Can a target $\mathbf{b}$ be produced?	$A\mathbf{x} = \mathbf{b}$ is consistent; equivalently $\mathbf{b} \in \text{Span}\{\mathbf{a}_i\}$.
Is every target in the codomain produced?	Pivot in every row; the columns span the codomain; T is onto.
Is a preimage unique whenever it exists?	$A\mathbf{x} = \mathbf{0}$ has only $\mathbf{0}$; pivot in every column; T is one-to-one.
How is T reconstructed?	$A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)]$, or use the images of any basis.

Three habits to keep

- (1) Separate existence from uniqueness.
- (2) Translate vector questions into pivots and homogeneous solutions.
- (3) Apply linearity to complete linear combinations.